

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 152, 146, 163, 163, 161, 154, 167, 156, 171]

Tap amplitudes (pixels): [127.94139283281233, 122.80065146407001, 126.21014222319853, 94.5568612000208, 111.07204868912791, 117.6860229593982, 95.1892851112981, 88.3911760301898, 94.57801012920498, 102.45974819410792, 78.39005038906404]



Based on the extracted features provided, here is the analysis and scoring for the MDS-UPDRS Item 3.4 Finger Tapping test.

Analysis of Extracted Features

1. Rhythm and Speed (Periods)

- **Data:** [150, 152, 146, 163, 163, 161, 154, 167, 156, 171]
- **Observation:** The tapping speed shows a gradual slowing trend. The subject starts with periods around 150 frames and ends with periods reaching 171 frames.
- **Interruptions:** There are no distinct outliers or massive spikes in the period data that would indicate a "halt" or "freeze" (arrest in movement). The rhythm is relatively continuous, though it slows down by approximately 14% from start to finish.
- **Clinical Correlate:** This matches the criteria for **slight slowing**.

2. Amplitude (Decrements)

- **Data:** [127.9, 122.8, 126.2, 94.6, 111.1, 117.7, 95.2, 88.4, 94.6, 102.5, 78.4]

- **Observation:**
 - **Taps 1–3:** The subject maintains a high amplitude (avg ~125 pixels).
 - **Taps 4–6:** There is a dip at tap 4, but the subject recovers amplitude at taps 5 and 6 (reaching ~117 pixels).
 - **Taps 7–11:** A sustained decrement becomes apparent. The amplitude drops consistently below 100 pixels (except a slight bump at tap 10), ending on the lowest value of 78.4 pixels.
- **Clinical Correlate:** While there is variability in the middle, the definitive, sustained decrement (fatigue) occurs in the latter half of the sequence (taps 7–11). This fits the description of **amplitude decrementing near the end** of the 10 taps.

Scoring Conclusion

The subject demonstrates **slight slowing** (periods increasing from ~150 to ~170) and **amplitude decrement near the end** (falling from ~125 range to ~80-90 range in the final taps). There are no hesitations or halts severe enough to warrant a higher score.

This profile aligns with **Score 1 (Slight)**:

- *1b) Slight slowing*
- *1c) The amplitude decrements near the end of the 10 taps*

Score: 1